

ITAI 3375 — Curating Data for AI Optimization

LAB 07 — Curate a Mini LLM Pre-Training Dataset

Module 07: Data for Foundation Models & LLMs — Pre-Training, Instruction Tuning & Evaluation

SLOs Assessed: 2, 4, 5 | Submission Format: This completed document + notebook + dataset artefacts

Student Information

Student Name:		
Date Submitted:		Section / CRN:

Lab Overview

This lab maps directly to Module 07 SLOs:

- SLO 2 — Apply the data curation lifecycle to assemble a pre-training corpus with documented filtering decisions
- SLO 4 — Design instruction-tuning datasets in correct JSONL formats with quality scoring
- SLO 5 — Construct evaluation data, apply contamination checks, and write professional data cards

You will curate a domain-specific mini LLM dataset end-to-end: define the scope, source and filter text, apply deduplication, build an instruction tuning set, perform contamination checks, and write a complete data card — exactly the artefacts a production LLM team would produce.

Part A — Domain Definition & Token Budget

Objective: Define the scope of your mini pre-training corpus before acquiring any data. A clear domain definition drives every subsequent filtering decision.

A1. Domain Selection

Select ONE domain from the options below, or propose your own (requires approval). This is the topic your mini-LLM would specialise in.

- a) Medical / Biomedical literature — PubMed abstracts, clinical guidelines, medical textbooks

- b) Legal documents — US case law summaries, contracts, regulatory text (Regulations.gov)
- c) Python programming — Python documentation, tutorials, Stack Overflow, GitHub README files
- d) Financial analysis — SEC filings (EDGAR), earnings call transcripts, financial news
- e) Scientific papers — arXiv abstracts in a single field (CS/AI, physics, biology, etc.)
- f) [] My own domain (describe fully): _____

A2. Domain Definition Document

Before searching for data, write precise inclusion/exclusion criteria. These become your quality filter guidelines.

Domain name (1 sentence)	
Target AI task(s)	e.g., question answering, summarisation, code completion
Inclusion criteria	What text BELONGS in this corpus? Be specific about topics, genres, time range
Exclusion criteria	What must be excluded? (off-topic, low-quality, harmful, non-English if targeting English model)
Language(s)	
Date range	e.g., 2018–2024, or 'any'
Min document length	e.g., ≥ 100 tokens, ≥ 3 paragraphs

A3. Token Budget Plan

Target: 100,000 tokens in your final corpus. Plan how you will reach this across at least 2 sources.

Data Source	Access Method	Target Tokens	License (SPDX)
TOTAL		Target: 100K	

Part B — Data Acquisition & Quality Filtering

Objective: Download from your chosen sources, apply the quality filtering pipeline, and document every filtering decision with before/after counts.

B1. Raw Acquisition Log

Source	Raw Document Count	Raw Token Count	Download Date

B2. Filtering Pipeline — Stage-by-Stage

Implement at least 4 of the 8 quality filtering stages from Module 07, Slide 06. For each stage applied, record the before/after document count and the threshold used.

Filter Stage	Tool / Implementation	Docs Before	Docs After	Threshold / Rule
Language identification (fastText LID)				
Boilerplate / HTML removal (trafilatura)				
Quality heuristics (word length, symbol ratio, etc.)				
Repetition filter (n-gram ratio)				
Perplexity filter (KenLM score)				
Toxicity / NSFW filter				
MinHash deduplication (datasketch)				
Other (describe):				

B3. Final Corpus Statistics

Final document count	
Final token count	Measure with tiktoken or HuggingFace tokeniser
Tokeniser used	e.g., cl100k_base (GPT-4), llama (Llama 3)
% retained after all filters	$(\text{final} / \text{raw}) \times 100$
Most aggressive filter	Which stage removed the most documents?
Mean document length	Tokens per document, mean $\pm$ std dev

Part C — Instruction Tuning Dataset

Objective: Create 50 instruction-response pairs in chat JSONL format. Achieve diversity across at least 5 distinct task types and quality-score every example.

C1. Task Type Distribution Plan

Before writing examples, define your 5+ task categories. Achieving diversity at planning stage prevents writing 50 variants of the same question.

Task Type	Brief Description + Example Instruction	Target Count

Task Type	Brief Description + Example Instruction	Target Count
TOTAL	TOTAL	TOTAL

C2. JSONL Format Confirmation

Write ONE complete example from each task type in the code block below using the Chat format (system / user / assistant roles). Include at least 3 of your 5+ categories here.

C3. Quality Scoring

Score ALL 50 examples on each dimension (1–5). Summarise the distribution here. Any example scoring < 3 on any dimension must be revised or replaced.

Quality Dimension	Score 5 (count)	Score 4 (count)	Score 3 (count)	Score 1–2 (revised)
Accuracy (is the response factually correct?)				

Quality Dimension	Score 5 (count)	Score 4 (count)	Score 3 (count)	Score 1–2 (revised)
Helpfulness (does it fully address the instruction?)				
Safety (no harmful, biased, or toxic content)				
Format (appropriate structure for the task type)				

Part D — Contamination Check

Objective: Verify that none of your 50 instruction examples appear in your pre-training corpus. Report contamination rate and remediation.

D1. Contamination Check Methodology

Method 1 used	☐ Exact substring match ☐ SHA-256 hash match ☐ Both
Method 2 used	☐ 8-gram overlap (compute token-level overlap between each instruction and corpus)
Tool / library used	e.g., Python str.find(), MinHash (datasketch), custom n-gram script
Overlap threshold for flagging	e.g., flag if > 30% of 8-grams in the instruction appear in training corpus

D2. Contamination Results

Total instruction examples checked	50
Exact matches found	Count — describe what matched and the source document
Near-duplicate matches (n-gram)	Count and average overlap percentage
Final contamination rate	$(\text{flagged} / 50) \times 100 = \underline{\hspace{1cm}}\%$

Examples removed / replaced	Count and brief description of what was removed
Final clean instruction set size	Must be ≥ 45 after remediation

Reflection: Why does contamination between instruction data and pre-training corpus matter for LLM evaluation validity?

Part E — Data Card

Objective: Write a complete, professional data card documenting your dataset for public release. A data card is the primary artefact that makes a dataset trustworthy.

E1. Pre-Training Corpus Data Card

Basic Information

Dataset Name	e.g., MedText-100K-v1.0
Version	v1.0.0 (SemVer)
Curated by	Your name, institution, date
License	SPDX identifier(s) — must cover all constituent sources

Composition

Total documents	
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Total tokens	
Language(s)	
Date range of source content	
Source breakdown	Source 1: N docs (N tokens) Source 2: N docs (N tokens)
Token distribution	Describe mean, median, min, max document length in tokens

Filtering & Processing

Describe every filter applied in the order applied. Be precise enough that another engineer could reproduce your pipeline:

Known Limitations & Biases

Describe: (a) what topics are underrepresented, (b) any known quality issues not fully addressed by your filters, (c) any demographic or temporal bias introduced by your source selection:

Intended Use & Prohibited Uses

Intended use	What AI tasks is this corpus suitable for?
Prohibited uses	What should this corpus NOT be used for? (e.g., high-stakes clinical decisions without human oversight)

Commercial use permitted?	☐ Yes ☐ No ☐ Conditional — explain
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E2. Instruction Tuning Set Data Card (Abbreviated)

Dataset Name	e.g., MedInstruct-50-v1.0
Example count (final)	
Task type distribution	Task 1: N Task 2: N Task 3: N ...
Quality scoring method	Describe how you scored and what threshold you used to pass/fail
Human review performed?	☐ Yes — by whom? ☐ No — explain quality assurance approach used instead
Contamination check result	Method, contamination rate, examples removed
Format	Chat JSONL — compatible with: HuggingFace TRL SFTTrainer, axolotl, LLaMA-Factory

Part F — Reflection & Critical Analysis

Objective: Apply critical thinking to your curation decisions and connect them to real-world LLM data engineering practice.

F1. The Quality vs. Quantity Trade-off

The LIMA paper (Zhou et al., 2023) showed 1,000 carefully curated examples outperformed 52,000 noisy ones for instruction tuning. Based on your experience building these 50 examples, do you believe this finding? What evidence from your own quality scoring data supports or challenges it?

F2. Scaling Challenge Analysis

You built a 100K-token corpus manually. Llama 3 used 15 trillion tokens. Identify the 3 biggest engineering challenges that arise when scaling YOUR pipeline from 100K to 15T tokens — and describe how each would need to be addressed (tools, infrastructure, processes).

F3. Constitutional AI Application

Review your 50 instruction examples. Identify 2 examples where your response could be made safer or more aligned with a constitutional principle. Write the original response and a revised version, and state which principle guided the revision.

#	Original Response (excerpt)	Revised Response (excerpt)	Constitutional Principle Applied
1			
2			

Grading Rubric

Component	Criteria for Full Credit	Points
Part A — Domain & Token Budget	Domain definition complete with inclusion/exclusion criteria; token budget	10

Component	Criteria for Full Credit	Points
	table filled with ≥ 2 sources; licenses specified	
Part B — Acquisition & Filtering	Raw acquisition log complete; ≥ 4 filter stages documented with before/after counts; final corpus statistics accurate	20
Part C — Instruction Tuning Set	Task type plan shows ≥ 5 categories; 3 examples in correct JSONL format; all 50 examples scored on 4 dimensions	25
Part D — Contamination Check	Both exact and n-gram methods attempted; contamination rate reported; reflection demonstrates understanding of why it matters	15
Part E — Data Card	Pre-training card complete across all 5 sections; instruction card covers 7 required fields; limitations are honest and specific	20
Part F — Reflection	LIMA reflection references specific quality score data; scaling analysis identifies real engineering challenges with solutions; CAI revision is substantive	10
Total		100

Instructor feedback: